LIM KER YEW

Details

Date of Birth:
6th August 2002

Nationality:
Malaysian

Location:
Sarawak, Malaysia

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Email:
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Profiles:
Lim Ker Yew
Isafe-nk

Languages

English, Malay, Mandarin

Programming

Python, C, Java, SQL, JavaScript, Dart

Frameworks

TensorFlow, Scikit-learn, XGBoost, OpenCV, React.js, Node.js, Flutter, Streamlit

Tools

Git, Docker, Android Studio, Microsoft SQL Server, SQLite

Summary

Computer Science with Artificial Intelligence graduate from the University of Nottingham Malaysia with a focused interest in Generative AI, intelligent workflow automation, and financial technology. Hands-on experience architecting autonomous agentic workflows, local RAG pipelines, and fintech solutions. Strong foundation in scalable web application development, designing data-driven backend architectures that integrate technologies such as MCP, n8n, and vector databases to address complex business challenges. Former President of the University Investment Club, bringing initiative, adaptability, and a strong sense of ownership to both leadership and collaborative engineering environments. Committed to delivering meaningful, high-impact results.

Education

BSc (Hons) Computer Science with Artificial Intelligence
University of Nottingham, Malaysia | Second Upper Class *2021-2025*

Cultural Exchange Program
Work & Travel, United States of America *2024*

Foundation in Science
University of Nottingham, Malaysia | CGPA: 3.75/4.00 *2020-2021*

Malaysia Certificate of Education (SPM)
SMK Lutong, Sarawak | Result: 7A, 3B+ *2015-2019*

Skill Highlights

- Project & Team Management
- Problem Solving & Decision Making
- Self motivated & Innovative
- Leadership drive & Optimistic mindset
- Financial Data Modeling & Analysis
- Generative AI & RAG pipelines
- Intelligent Workflow Automation
- Web Application Development
- Cloud Deployment & Version Control
- API Architecture & Deployment

Portfolio

AI-Based Risk Assessment & Portfolio Diversification *Final Year Project*
Local Agentic AI File Explorer *Personal macOS Project*
Real-Time Eye-Tracking Mobile App *Computer Vision App*
Autonomous Robot Pathfinding Simulation *Algorithm Simulation*
Real-Time Train Scheduling & Journey Management API *Backend Service*
Full-Stack E-Commerce Web Application *Course Project*

Experience

Sales Associate at Surf Style Retail (USA) *2024*

L'Oreal Brandstorm Malaysia *2023*
National Top 3 Finalist

WorldQuant Brain Alphathon *2023*
Rank 9 (Gold Level)

President of Investment Club at University of Nottingham Malaysia *2021-2022*

Web Developer Intern at Fusionex (Remote) *2022*

Course Consultant Assistant at Finspark (Remote) *2021*

School Prefect of SMK Lutong *2015-2019*

Karate Shitokai Brunei Open *2018*
1 Gold, 1 Bronze

15th "Peer Cup" Hong Kong Wushu International Championship *2017*
2 Gold, 1 Silver

Significant Projects Details

AI-Based Risk Assessment and Portfolio Diversification in Stocks *(Final Year Project)*

- Architected and deployed a full-stack hybrid robo-advisory platform using Python and Streamlit, integrating the Financial Modeling Prep API to retrieve real-time market data, company profiles, and discounted cash flow (DCF) values.
- Engineered an adaptive investor profiling engine that seamlessly transitions from a deterministic rule-based algorithm to a Scikit-learn Random Forest Regressor, processing demographic and behavioral data to predict continuous risk scores with an R^2 of 0.91.
- Developed a machine learning stock screener by training a Random Forest Classifier on historical S&P 500 data, utilizing synthetic oversampling and technical indicators (RSI, MACD, Bollinger Bands, and Moving Averages) to categorize equities into dynamic risk tiers.
- Designed a valuation-weighted portfolio allocation algorithm that dynamically matches user risk profiles with customized equity recommendations, enforcing automated rebalancing protocols to prevent sector overconcentration and ensure robust diversification.

Local Agentic AI File Explorer *(Personal Project)*

- Architected a privacy-first, local AI file explorer utilizing a custom FastAPI backend and Ollama on Apple Silicon, enabling semantic search and file manipulation with zero external cloud dependencies.
- Engineered an automated data observability pipeline using Python's watchdog daemon and asynchronous background threading (asyncio) to dynamically monitor local directories, utilizing timestamp deduplication to update a Qdrant vector database in real-time.
- Implemented LLM function calling (Agentic AI) to bridge natural language queries with native macOS file operations, allowing a local Llama 3 model to autonomously execute Python system commands.
- Developed a decoupled, responsive React frontend using Vite and Tailwind CSS to visualize live local directory trees and stream real-time AI responses via custom REST API endpoints.

Real-Time Eye-Tracking Mobile App *(Software Engineering Coursework)*

- Engineered a native Android cognitive assessment application in Java, integrating Google ML Kit's Face Detection API to perform real-time facial landmark identification and evaluate eye-open probabilities.
- Architected a real-time computer vision pipeline utilizing OpenCV to isolate and track user irises through a sequence of grayscale conversion, adaptive thresholding, morphological operations, and Canny edge detection.
- Implemented spatial tracking algorithms using image moments to calculate the centroid of iris contours for precise gaze detection, seamlessly exporting analyzed reaction time and accuracy data into formatted Excel reports.

Autonomous Robot Pathfinding Simulation *(Algorithm Simulation)*

- Implemented BFS and Dijkstra algorithms with predecessor path reconstruction to calculate optimal routes through grid environments while dynamically avoiding complex obstacles.
- Built an interactive Pygame GUI to visualize real-time algorithm execution, displaying performance metrics like execution time and path length for comparative analysis.
- Engineered dynamic obstacle generation with algorithm-based pathability verification to ensure valid route existence and manual obstacle manipulation.

Experience Details

Cultural Exchange Program, USA *(June 2024 - Oct. 2024)*
Sales Associate at Surf Style Retail

- Executed daily store operations, managed high-volume inventory, and handled cashier duties in a fast-paced retail environment.
- Adapted quickly to a multicultural workplace, communicating effectively with international teams and customers to solve problems on the floor.

Investment Club, University of Nottingham *(Sept. 2022 - Apr. 2023)*
President

- Directed the executive committee to organize practical workshops and networking events, actively promoting financial literacy across the student body.
- Partnered with industry professionals to host technical sessions, giving students direct insight into real-world investment strategies and market trends.

Fusionex, Remote *(June 2022 - Aug. 2022)*
Web Developer

- Built and shipped scalable web application features using React.js and Node.js, directly improving platform functionality and the end-user experience.
- Worked closely within an 8-person cross-functional engineering team to deliver code on tight deadlines, actively participating in peer reviews to maintain code quality.

Finspark, Remote *(Apr. 2021 - Aug. 2021)*
Course Consultant Assistant

- Led the front-end development and UI redesign of the main course website, making the platform more accessible and easier to navigate for users.
- Digitized and structured core finance materials into the new system, accelerating the onboarding process for new students.